Optima Academy Online
An Education Experience Company

Teacher Manual — Grade 6 Mathematics

FACILITATOR GUIDE · Florida course 1205010 · Full year · 9 modules

Facilitator copy. Contains grading guidance, answer-guide logic, and standard codes. **Do not publish to students.** This document is not part of the student-facing Canvas course.

This manual documents the Grade 6 Mathematics course as it is actually built (nine modules of lesson pages, prior-knowledge checks, independent practice, mid-module checks, module summatives, and VR components), and explains how one facilitator supports roughly 500 asynchronous students inside it. Where a detail could not be confirmed from the built course, it is flagged **[VERIFY]**. Two items were added at the course team's request and are marked with a gold ADDED tag: a **live 4-day/week adaptation** (Section 11) and a **standards coverage overview** (Section 12).

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1. How this course works

A TYPICAL LESSON

Each lesson lives on a Canvas page dressed in the Optima navy/gold chrome (navy header with the owl, a metadata strip, a lavender title strip, gold-kicker content cards, navy footer). Reading top to bottom, a student meets:

- **Prior Knowledge Check** — a short prerequisite-review quiz that surfaces what a student needs from earlier lessons/grades before starting.
- **Exploring the Math** — the concept, worked through with embedded instructional video(s) and Wiris-rendered equations.
- **Vocabulary** and any **Essential Questions** cards.
- **VR Lesson Resource** — an immersive activity on Optima's VR campus that reinforces the concept.
- **Independent Practice** — the graded touchpoint: a Canvas quiz of ~4–10 problems with unlimited attempts and check-as-you-go feedback and hints.

The graded touchpoints are almost all auto-scored Canvas quizzes, so the student gets immediate feedback and the teacher spends no time scoring them.

A TYPICAL WEEK

Because the course is On-Demand, a "week" is defined by pacing toward the quarter's due dates rather than by fixed class meetings. A student's normal cycle:

SEGMENT	WHAT THE STUDENT DOES	GRADED?	TIER
Prior Knowledge Check	Warms up prerequisite skills for the lesson	Yes (auto)	Tier 1
Lesson + VR	Watches/reads the lesson; completes the VR Lesson Resource	VR: yes	Tier 4 / 2
Independent Practice	Works ~4–10 problems, checking each answer	Yes (auto)	Tier 1
Mid-Module Check	1–2 per module: a progress checkpoint	Yes (auto)	Tier 1
Summative Review	End-of-module review, unlimited attempts	Completion	Tier 1
Summative Assessment	End-of-module test, 2 attempts, graded for correctness	Yes (auto)	Tier 1

[VERIFY] As built, this course is **heavily Tier 1 (auto-graded)**. No structured discussions, teacher-graded essays, or Canvas survey-mode module wrappers were found in the export. That is faithful to the source — but it means the 60/25/15 balance is not yet present. Sections 2, 7, and 11 mark where a facilitator can add Tier 2/3 touchpoints (especially in a live section) if the course team wants them. Confirm the VR-activity grading method against Canvas before term.

MAJOR ASSESSMENTS

There is one summative pair per module. Because they auto-score, they cost the teacher no grading time; the teacher's job is to **read the results** and reteach.

ASSESSMENT	WHERE	WHAT IT ASKS	TEACHER TIME
Summative Review (×9)	End of each module	Full-module review, unlimited attempts, completion credit	~0 (auto)
Summative Assessment (×9)	End of each module	Graded for correctness, 2 attempts	~0 (auto) + review of data
VR Activities/Projects	Select lessons (e.g., 1.01 VR Activity)	An immersive task with a capture/submission	~2–4 min each, if teacher-scored **[VERIFY]**

2. Ungraded interactives & the module wrappers

Ungraded practice is formation, not busywork. The check-as-you-go feedback and hints inside every Independent Practice and the immersive VR Lesson Resources give students immediate feedback and cost you nothing to run. A student skipping them does not show up as a missing grade; it shows up indirectly in Mid-Module Check and Summative performance. Watch the checks, not the practice logs.

RECOMMENDED ADDITION

Module intro/outro wrappers (not yet in the course). Optima's model builds these as **Canvas Quizzes in survey mode** — not wiki pages — so responses become sortable cohort data. If the course team adds them, the highest-leverage teacher move in the whole system is to spend **15–20 minutes per module triaging outro responses**: capture a first vs. refined answer to the module's essential question, a 1–5 confidence rating (the early-warning signal), a "what's still unclear" field, and a virtue reflection. Cluster the low-confidence students and the repeated unclear concepts, then act with a targeted announcement or an individual message. Until wrappers exist, the Mid-Module Check item analysis is your best cohort-level signal.

3. AI & VR in this course (student-facing)

AI

The course's posture (stated in the syllabus): students **may use AI to support learning** — explaining a concept another way, checking understanding, brainstorming — but AI may never do the thinking or the work an assignment asks for, and any AI use should be briefly disclosed. Frame this as awareness, not a required tool, and never add new AI requirements to a lesson. The honest goal: a student should be able to judge whether an AI's answer is any good, which requires being able to do the math without it.

VR

Every lesson carries a **VR Lesson Resource**, and select lessons carry a **VR Activity/Project** with a capture or submission. Students join immersive sessions on Optima's VR campus to reinforce the concept. Describe VR to families by what the student does and produces; if a student cannot get into VR for a session, there is an alternative way to take part so they do not fall behind. Review VR captures in a batch, or assemble them into a gallery walk for a live section.

4. Using Ignite AI to support students (teacher-facing)

> **[VERIFY]** Ignite AI's exact features and availability vary by Canvas release and by what Optima has enabled. Confirm the current feature set in Optima's Canvas and describe capabilities functionally — don't promise a button that isn't turned on.

Used well, the Canvas AI assistant helps make 500:1 feel personal. In functional terms it can:

- **Summarize quiz & Mid-Module-Check analytics** to find the cohort's sticking points fast (e.g., "most of M4 missed unit-rate conversions").
- **Surface students who need a nudge** — declining Summative scores, long inactivity — and draft a warm check-in you then edit and own.
- **Generate practice variants** for a student who needs another rep on a Tier 1 skill.
- **Draft family communications** (clear, kind summaries) and translate when helpful.

The governing rule: AI drafts, the teacher decides. You read, edit, and own anything sent to a student or family. AI is never used to surveil, rank, or auto-penalize (see Section 8).

5. Grading categories & Canvas gradebook setup

Map assignment groups to the tier system. Suggested groups and weights, honoring the 60/25/15 emphasis where the course allows — but this course as built is Tier-1-dominant, so weights lean toward practice and summatives. **Confirm the actual point values in the built quizzes before finalizing.**

ASSIGNMENT GROUP	CONTAINS	METHOD	SUGGESTED WEIGHT **[VERIFY]**
Practice & Checks T1	Prior Knowledge Checks, Independent Practice, Mid-Module Checks	Auto / completion	40%
Module Summatives T1	Summative Review (completion) + Summative Assessment (correctness)	Auto	40%
VR & Activities T2	VR Lesson Resources, VR Activities/Projects	Completion / rubric	20%

Step-by-step Canvas setup

1. Gradebook → **Assignment Groups**: create the three groups above.
2. Gear menu → **Weight final grade by assignment group**; enter the weights.
3. Set the **grading scheme** (school standard).
4. Mark Summative Reviews and VR Lesson Resources as **Complete/Incomplete** (completion), Summative Assessments as **Points**.
5. Set **late/missing policies** to match the syllabus: no penalty for individual due dates within a quarter; end-of-quarter zero for incomplete work; 10% deduction in the final 14 days.
6. Use the **module-requirements / progress** view as your at-a-glance completion signal across the ungraded scaffolding.

6. Rubrics for every teacher-graded item

Almost everything in this course auto-scores and needs **no rubric**. Attach a native Canvas rubric only to items a human scores — currently the VR Activities, and any Tier 3 checkpoint a facilitator adds (see Section 11). Keep rubrics to 3–4 criteria, identical across modules, so scoring stays ~5 min/student.

STANDARD MODULE-ASSESSMENT / REASONING RUBRIC (FOR ANY TEACHER-SCORED MATH TASK)

CRITERION	FULL CREDIT	PARTIAL	NONE
Mathematical understanding	Correct concepts and results; method fits the problem	Right idea, minor errors	Concept not shown
Quality of reasoning	Steps are logical, complete, and justified	Gaps or unexplained steps	No reasoning shown
Clarity & precision	Clear notation, labels, units; easy to follow	Some imprecision	Unclear/unlabeled
Intellectual honesty	Names what's uncertain; shows real work, own thinking	Partial candor	Copied / AI-filler

VR ACTIVITY RUBRIC

CRITERION	FULL CREDIT	PARTIAL	NONE
Task completed in VR	All steps done; capture/submission present	Some steps missing	Not attempted
Math connection	Clearly links the activity to the lesson concept	Weak/implicit link	No connection
Effort & care	Sincere, thoughtful attempt	Rushed	Nonsense/off-task

Free-response items: if the course team turns any Summative item into a short-answer/free-response question, add a compact 2–3 point answer guide (what a full-credit answer must contain, what earns partial, what earns none). Keep these guides in this manual only — never shown to students.

7. Quizzes, activities & discussions — how to run & grade

TYPE	WHAT IT IS / HOW IT SCORES	RECOMMENDED TEACHER MOVE
Auto-graded quiz (PKC, Independent Practice, Mid-Module Check, Summative Assessment)	Canvas Quiz, auto-scored; unlimited attempts (checks/reviews) or 2 attempts (summatives)	Don't grade — **read the item analysis**. Reteach the 1–2 most-missed ideas via announcement.
Summative Review	Completion, unlimited attempts	Confirm completion in the gradebook; no scoring.
VR Activity/Project	Immersive task with capture/submission	Batch-review captures against the 3-criterion VR rubric; drop 2–3 public model shout-outs.
Structured discussion T2 (add for live)	Native Canvas Discussion with a specific prompt	Batch-read; score substantive/thin/missing; drop 2–3 public comments that model good reasoning.

RECOMMENDED ADDITION

One good math discussion prompt beats ten vague ones. If you add a Tier 2 discussion (valuable in a live section), make it specific: "*State one strategy from this module you'd use to check whether an answer is reasonable, and explain why it works. Then reply to one classmate: does their check actually catch a wrong answer? Strengthen it or show where it falls short.*"

8. Coach, not catch

Your job is to move each student forward, not to detect and punish. In practice:

- Feedback names the next step, not just the error; lead with what's working.
- Unlimited attempts on practice are a **feature** — re-attempts are how the skill forms, not a loophole.
- Use analytics and Ignite AI to **find students who need help early**, then reach out — not to build a case.

- On suspected AI misuse or copying: open a conversation, ask the student to talk through their thinking, treat it as formation. Avoid accusation-first workflows.
- Public comments (in VR gallery or discussion) model good thinking; they never single out weak work.

9. Weekly announcement scripts

For asynchronous students the weekly announcement is your main presence. Clone this template, then use the drafted per-module openers.

> **[Module N] — [week focus]**
> Scholars: this week you should be in **[Module N, Lessons x–y]**. The one thing not to miss: **[due summative / tricky concept]**. A tip: **[one specific strategy]**. Stuck? Bring it to study hall **[day/time/link]** — that's what it's for. Keep going; you've got this. — [Your name]

MODULE	DRAFTED OPENER (FILL THE BRACKETS)
M1	Welcome! We start with rational numbers, opposites, and absolute value — the number line is your best friend this module. Watch signs carefully.
M2	Operations with fractions & decimals. The classic trap is lining up decimals and flipping-to-multiply for division — slow down on setup.
M3	Factors, multiples, and exponents. GCF vs. LCM trips everyone up — ask "am I splitting into groups (GCF) or finding a shared cycle (LCM)?"
M4	Ratios, rates, and percentages — the most real-world module yet. Unit rate is the key that unlocks most problems.
M5	Integer operations and order of operations. Sign rules + PEMDAS together; a single wrong sign changes everything.
M6	Expressions, inequalities, and equations — the doorway to algebra. Translate words to symbols carefully and keep equations balanced.
M7	Data! Mean/median, box plots, histograms. Ask what a graph hides as much as what it shows.
M8	The full coordinate plane — all four quadrants now. Mind your signs when you plot and measure distance.
M9	Area, surface area, and volume — we close the year with shapes. Draw the net; it makes surface area obvious.

Situational scripts: (a) *Low-confidence check-in* — "If the last check felt shaky, that's normal — here's a 5-minute reset and my study-hall link." (b) *Pre-summative* — "The Module N Assessment opens [date]. Do the Summative Review first — it's unlimited attempts and it's the review." (c) *Re-engagement* — "Haven't seen work from you in a bit — reply and tell me where you're stuck; we'll get you moving."

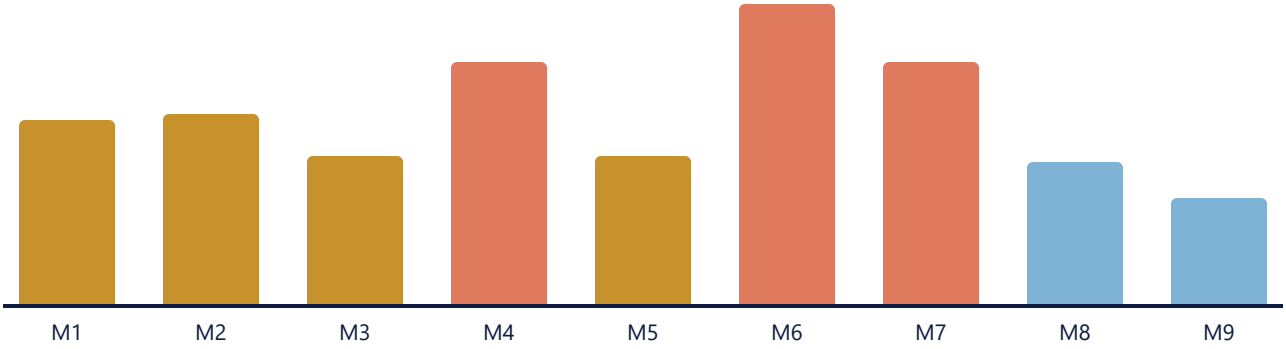
10. Teacher workflow & workload map

A normal week, in order

1. Scan Mid-Module-Check / Summative **item analysis** for the most-missed ideas (~20 min).

2. Batch-review any VR captures against the 3-criterion rubric.
3. Post the weekly announcement (reteach the 1–2 sticking points).
4. Answer messages; send 2–3 warm check-ins to inactive or declining students.

Because the graded core auto-scores, this course is **light on grading hours** — the teacher time goes into reading analytics, VR review, reteach announcements, and outreach. The bars below show **relative** attention by module (driven by module size: lesson + graded-item counts), not raw hours.



Blue = lighter, gold = moderate, coral = heavier. **Plan extra time on M4, M6, and M7** (most lessons and graded items). M9 is the lightest (2 lessons). Heights are proportional to per-module page counts in the build (M1 35, M2 36, M3 28, M4 46, M5 28, M6 57, M7 46, M8 27, M9 20).

11. Running this course live, 4 days/week ★ ADDED

ADDED AT REQUEST

The course is authored asynchronously; a live section uses the same Canvas content unchanged and simply adds a class rhythm on top of it. Below is a **repeatable weekly template** for a section meeting **four days a week**, plus a per-module pacing guide.

THE 4-DAY WEEKLY RHYTHM (USE THE CANVAS LESSON PAGES AS YOUR BOARD)

DAY	FOCUS	HOW THE ASYNC PIECES MAP ONTO THE LIVE MEETING
Day 1 Launch	Diagnose + introduce	Open with the lesson's **Prior Knowledge Check** as a warm-up (project it, or have students take it live) to see who's ready. Teach the concept from the **Exploring the Math** card; work its worked examples together. A short number talk makes a strong bell-ringer.
Day 2 Practice	Guided → independent	Do the first few **Independent Practice** problems together, then release students to finish while you circulate. The check-as-you-go feedback becomes your live formative signal — pull a small group around whoever the hints aren't reaching.
Day 3 Extend	Next lesson + check	Introduce the next lesson/concept and its practice. Where a **Mid-Module Check** falls, give it in class and review the item analysis on the spot so misconceptions don't harden.
Day 4 Apply & reteach	VR + small-group	Run the **VR Lesson Resource / VR Activity** together (assemble captures into a quick gallery walk). Use the rest for **targeted reteach** of the week's most-missed idea, and open the **Summative Review** as it comes due at module's end.

Weekly cadence rule of thumb: cover **~2–3 lesson segments per week**, one summative per module, and one VR touch per week. Keep Day 4 flexible — it absorbs reteach, VR, and summative days depending on where the module lands.

PER-MODULE LIVE PACING (FITS THE 2026–2027 CALENDAR'S ~36 WEEKS)

MODULE	LESSONS	SUGGESTED WEEKS	LIVE-WEEK NOTE
M1	1.01–1.03	~3 weeks	Number-line warm-ups daily; sign errors are the theme.
M2	2.01–2.04	~3–4 weeks	4 lessons — one per ~4-day week.
M3	3.01–3.03	~2–3 weeks	GCF/LCM game on Day 4.
M4	4.01–4.05	~4–5 weeks	**Heavy** — 5 lessons; budget a full week for percentages.
M5	5.01–5.03	~3 weeks	Integer chips/number line manipulatives on Day 1.
M6	6.01–6.05	~5 weeks	**Heaviest** — the algebra doorway; add a Tier 2 "translate the words" discussion.
M7	7.01–7.05	~4–5 weeks	**Heavy** — a live data-collection project is a natural Tier 3 checkpoint.
M8	8.01–8.03	~2–3 weeks	Graph-paper stations; coordinate battleship on Day 4.
M9	9.01–9.02	~2 weeks	**Light** — hands-on nets and boxes to close the year.

TWO LIVE-ONLY TOUCHPOINTS WORTH ADDING

- **A weekly number talk or short discussion (Tier 2)** — 10 minutes of "how did you see it?"; in Canvas, a specific-prompt discussion (Section 7).
- **One Tier 3 checkpoint per module** — turn one summative task into a short written-reasoning item graded on the Section 6 rubric, so students explain *why*, not just compute. Keep it to one per module to stay inside the time budget.

12. Standards coverage map (CPALMS → modules) ★ ADDED

ADDED AT REQUEST

Course: **M/J Grade 6 Mathematics (#1205010)**, Florida B.E.S.T. Standards. The map below is reconstructed from the module and lesson titles against the CPALMS benchmark strands (standards do not appear on student pages).

[VERIFY] This alignment is inferred from titles + strand codes; confirm it against the course team's official standards alignment before using it for reporting. Codes are teacher-copy only.

BY MODULE (QUICK REFERENCE)

MODULE	TOPIC	BENCHMARKS ADDRESSED
M1	Integers, Rational Numbers & Absolute Value	MA.6.NSO.1.1, 1.2, 1.3, 1.4
M2	Operations on Fractions & Decimals	MA.6.NSO.2.1, 2.2, 2.3
M3	Factorization, Composition & Exponents	MA.6.NSO.3.1, 3.2, 3.3, 3.4
M4	Ratios, Rates, Percentages & Conversions	MA.6.NSO.3.5; MA.6.AR.3.1, 3.2, 3.3, 3.4, 3.5
M5	Integers — Operations & Order of Operations	MA.6.NSO.4.1, 4.2 (previews MA.6.AR.1.3)
M6	Algebraic Expressions, Inequalities & Equations	MA.6.AR.1.1, 1.2, 1.3, 1.4; MA.6.AR.2.1, 2.2, 2.3, 2.4
M7	Collecting, Using & Interpreting Data	MA.6.DP.1.1, 1.2, 1.3, 1.4, 1.5, 1.6
M8	The Coordinate Plane	MA.6.GR.1.1, 1.2, 1.3
M9	Area, Surface Area & Volume	MA.6.GR.2.1, 2.2, 2.3, 2.4

THREADED THROUGH EVERY MODULE

The Mathematical Thinking & Reasoning and literacy standards are not tied to one module — they run throughout (in the practice, VR reasoning tasks, and explanations):

- **MA.K12.MTR.1.1–7.1** — effortful learning; multiple representations; fluency; discussion/justification; patterns & structure; reasonableness of solutions; real-world application.
- **ELA.K12.EE.1.1–6.1** — cite evidence/justify reasoning; comprehend; infer; collaborate; follow format rules; appropriate voice.
- **ELD.K12.ELL.MA.1** and **ELD.K12.ELL.SI.1** — English-language development in mathematics and for social/instructional purposes.

BY STRAND (WHERE EACH BENCHMARK IS ASSESSED)

STRAND	BENCHMARKS	TAUGHT & ASSESSED IN
NSO Number Sense & Operations	1.1–1.4 · 2.1–2.3 · 3.1–3.5 · 4.1–4.2	M1–M5
AR Algebraic Reasoning	1.1–1.4 · 2.1–2.4 · 3.1–3.5	M4, M6
DP Data Analysis & Probability	1.1–1.6	M7

STRAND	BENCHMARKS	TAUGHT & ASSESSED IN
GR Geometric Reasoning	1.1–1.3 · 2.1–2.4	M8, M9

13. What makes this course classical & cultivating culture

Classical moves in this course: mathematics is taught as reasoning, not recipe. The prior-knowledge checks honor that understanding builds on understanding; the "Exploring the Math" cards move from concrete models and number lines to abstract procedure (the MTR arc); real-world contexts ground each concept in the world; and the emphasis on checking reasonableness and justifying method treats a student as a young mathematician accountable to truth, not just to an answer key. The VR activities invite wonder before mechanism — students *experience* a concept before formalizing it.

Cultivating culture asynchronously: the teacher builds real community by naming students' good thinking in public shout-outs and gallery walks, modeling curiosity in weekly announcements ("here's a slicker way three of you found"), and treating feedback as relationship. Hold the virtue thread visible — perseverance through a hard problem set, honesty in showing real work — and the course becomes formation, not just a sequence of quizzes.

14. Sensitive-content flags & family notes

This is a mathematics course; it contains **no mature, violent, or contested material** that families would need advance notice of. Two ordinary items are worth a proactive note:

- **VR headset use.** Tell families up front that lessons include immersive VR, roughly how long a session runs, and that there is a non-VR alternative if a student can't get into a session. (Editable script:) *"This course includes short weekly VR activities on Optima's campus. If your scholar can't get into VR for a session, they won't fall behind — there's an alternative way to take part. Reach out any time with questions. — [Your name]"*
- **Math anxiety / pacing.** Some students arrive anxious about math. The unlimited-attempt design is your ally — say so to families, and frame re-attempts as how the skill forms (Section 8).

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